Multi-area OSPF configuration

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Purpose:

The purpose of this lab is to familiarize myself with the configuration of backbone routing, where OSPF (Open Shortest Path First) is a common, and specifically multi area OSPF which is better adapted to large networks by simplification of calculation. Also, to learn how to operate a multilayer switch to operate in layers 2 and 3 of the TCP/IP stack to improve network simplicity and performance.

Background Information:

OSPF:

Open Shortest Path First (OSPF) is a routing protocol based on link-state routing, designed to find the shortest path for packets between routers. It is very scalable and efficient, particularly noticeable when used in large networks divided into multiple areas such as our setup. I configure OSPF for both IPv4 and IPv6 networks, focusing on a multi-area setup.

OSPF, as a Link-State Routing Protocol, operates by exchanging topology information between routers to construct the network's Link State Database (LSDB). Unlike distance-vector protocols like RIP, OSPF converges more quickly and can be divided into multiple areas for scalability. In single-area OSPF, all routers share the same LSDB, which can lead to inefficiency in large networks. Multi-area OSPF solves this by breaking the network into areas, with Area 0 (the backbone) connecting all other areas. This division reduces the size of the LSDBs routers need to manage and improves scalability. OSPFv2 supports IPv4, while OSPFv3 is used for IPv6, both of which maintain the same concepts of areas, LSAs, and a hierarchical design.

At the core of OSPF is Dijkstra’s Algorithm, which calculates the Shortest Path First (SPF) to determine the best routes. The algorithm constructs a Shortest Path Tree (SPT), where routers form nodes and links form edges, each with an associated cost. The router starts by setting the cost to itself as zero and infinity

for all other routers, then explores its neighbors, updating their costs if shorter paths are found. This process repeats until all routers have been visited and the shortest paths calculated. OSPF also runs this algorithm to update its routing table when it detects network changes.

In a multi-area OSPF setup, the algorithm operates separately in each area, ensuring that most routers only must calculate paths within their local area to limit overhead. Area Border Routers (ABRs) manage inter-area routing by summarizing routes between areas.

Multi-Layer Switch:

Adding a multi-layer switch to the topology improves switching and routing capabilities. Multi-layer switches, which can operate at both Layer 2 (switching) and Layer 3 (routing) can participate in OSPF by enabling routing on the switch's interfaces, integrating into the OSPF network to forward traffic based on routing protocols while still providing switching for traffic within their network. In a multi-area OSPF topology, the multi-layer switch can function as an internal router or ABR, in my case an ABR, helping to route traffic between different OSPF areas.

Summary:

In this lab I configure 3 separate OSPF areas. Areas 0, 1, and 2. Area 0 consists of 2 network devices, two ABRs, one is a true router, bridging area 0 and 1 while the other is a multi-layer switch bridging from area 0 to area 2. Area 1 has two routers, with the second connecting to an end user. Area 2 has 2 routers, and has an end user after the second router, all devices are fully able to communicate in both IPv4 and IPv6 and cross network traffic is routed using OSPFv3 and OSPFv2 for IPv6 and IPv4 respectively.

Commands:

Switch

ip routing – allows the switch to route packets, virtually making it a router/layer 3 switch
sdm prefer dual-ipv4-and-ipv6 route – allows the switch to use ipv4 and ipv6 in routing
no switchport – allows the port to have an ip address and have the port act like a router's

port ip ospf 10 area 0 – sets up ospf on the port by setting the ospf area and process id for ipv4 ipv6 ospf 10 area 0 – sets up ospf on the port by setting the ospf area and process id for ipv6
router ospf 10 – makes the ospf process id 10
router-id 5.5.5.5 – makes the ospf router id 5.5.5.5
network 192.168.6.0 0.0.0.255 area 0 – allows the ospf network to be set up for the 192.168.6.0/24 network 192.168.0.0 0.0.0.255 area 0 – allows the ospf network to be set up for the 192.168.0.0/24 network 192.168.4.0 0.0.0.255 area 2 – allows the ospf network to be set up for the 192.168.4.0/24

Router 0

ipv6 unicast routing – allows for ipv6 addresses to be used on ports ip ospf 10 area 1 – allows an interface to be configured with ospf process id and area for ipv4 ipv6 ospf 10 area 1 – allows and interface to be configured with ospf process id and area for ipv6 router ospf 10 – sets router process id router-id 1.1.1.1 – sets the router id for ospf to determine the DR and BDR network 192.168.1.0 0.0.0.255 area 1 – allows the network 192.168.4.0/24 to be routed through ospf network 192.168.0.0 0.0.0.255 area 0 – allows the network 192.168.5.0/24 to be routed through ospf ipv6 ospf 10 – sets the ipv6 ospf process id

Router 1

ipv6 unicast routing – allows for ipv6 addresses to be used on ports ip ospf 10 area 1 – allows an interface to be configured with ospf process id and area for ipv4 ipv6 ospf 10 area 1 – allows and interface to be configured with ospf process id and area for ipv6 router ospf 10 – sets router process id router-id 3.3.3.3 – sets the router id for ospf to determine the DR and BDR network 192.168.1.0 0.0.0.255 area 1 – allows the network 192.168.4.0/24 to be routed through ospf network 192.168.2.0 0.0.0.255 area 1 – allows the network 192.168.5.0/24 to be routed through ospf ipv6 ospf 10 – sets the ipv6 ospf process id

Router 2

ipv6 unicast routing – allows for ipv6 addresses to be used on ports ip ospf 10 area 1 – allows an interface to be configured with ospf process id and area for ipv4 ipv6 ospf 10 area 1 – allows and interface to be configured with ospf process id and area for ipv6 router ospf 10 – sets router process id router-id 4.4.4.4 – sets

the router id for ospf to determine the DR and BDR network 192.168.2.0 0.0.0.255 area 1 – allows the network 192.168.4.0/24 to be routed through ospf network 192.168.3.0 0.0.0.255 area 1 – allows the network 192.168.5.0/24 to be routed through ospf ipv6 ospf 10 – sets the ipv6 ospf process id

Router 3

ipv6 unicast routing – allows for ipv6 addresses to be used on ports ip ospf 10 area 2 – allows an interface to be configured with ospf process id and area for ipv4 ipv6 ospf 10 area 2 – allows and interface to be configured with ospf process id and area for ipv6 router ospf 10 – sets router process id router-id 2.2.2.2 – sets the router id for ospf to determine the DR and BDR network 192.168.4.0 0.0.0.255 area 2 – allows the network 192.168.4.0/24 to be routed through ospf network 192.168.5.0 0.0.0.255 area 2 – allows the network 192.168.5.0/24 to be routed through ospf ipv6 ospf 10 – sets the ipv6 ospf process id

Router 4

ipv6 unicast routing – allows for ipv6 addresses to be used on ports ip ospf 10 area 2 – allows an interface to be configured with ospf process id and area for ipv4 ipv6 ospf 10 area 2 – allows and interface to be configured with ospf process id and area for ipv6 router ospf 10 – sets router process id router-id 1.1.1.1 – sets the router id for ospf to determine the DR and BDR network 192.168.6.0 0.0.0.255 area 2 – allows the network 192.168.6.0/24 to be routed through ospf network 192.168.5.0 0.0.0.255 area 2 – allows the network 192.168.5.0/24 to be routed through ospf ipv6 ospf 10 – sets the ipv6 ospf process id

Configurations:

Router0:

```
Current configuration : 1668 bytes
Last configuration change at 21:54:21 UTC Mon Sep 30 2024
version 16.9
service timestamps debug datetime msec
service timestamps log datetime msec
platform qfp utilization monitor load 80
platform punt-keepalive disable-kernel-core
hostname Router0
```

```
boot-start-marker
boot-end-marker
vrf definition Mgmt-intf
address-family ipv4
exit-address-family
address-family ipv6
exit-address-family
no aaa new-model
login on-success log
subscriber templating
ipv6 unicast-routing
multilink bundle-name authenticated
license udi pid ISR4321/K9 sn FLM2408005M
no license smart enable
diagnostic bootup level minimal
spanning-tree extend system-id
redundancy
mode none
interface GigabitEthernet0/0/0
ip address 192.168.0.1 255.255.255.0
ip ospf 10 area 0
negotiation auto
ipv6 address FE80::1 link-local
ipv6 address 2001:DB8:ACAD:A::1/64
ipv6 enable
ipv6 ospf 10 area 0
interface GigabitEthernet0/0/1
ip address 192.168.1.1 255.255.255.0
ip ospf 10 area 1
negotiation auto
ipv6 address 2001:DB8:ACAD:B::1/64
ipv6 enable
ipv6 ospf 10 area 1
interface GigabitEthernet0/2/0
no ip address
negotiation auto
interface GigabitEthernet0/2/1
no ip address
negotiation auto
interface GigabitEthernet0
vrf forwarding Mgmt-intf
no ip address
negotiation auto
router ospf 10
router-id 6.6.6.6
ip forward-protocol nd
no ip http server
ip http secure-server
ip tftp source-interface GigabitEthernet0
ipv6 router ospf 10
router-id 6.6.6.6
control-plane
```

```
line con 0
transport input none
stopbits 1
line aux 0
stopbits 1
line vty 0 4
login
end
```

Router1:

```
Current configuration : 1593 bytes
Last configuration change at 21:47:00 UTC Mon Sep 30 2024
version 16.9
service timestamps debug datetime msec
service timestamps log datetime msec
platform qfp utilization monitor load 80
platform punt-keepalive disable-kernel-core
hostname Router1
boot-start-marker
boot-end-marker
vrf definition Mgmt-intf
address-family ipv4
exit-address-family
address-family ipv6
exit-address-family
no aaa new-model
login on-success log
subscriber templating
ipv6 unicast-routing
multilink bundle-name authenticated
license udi pid ISR4321/K9 sn FLM24060912
no license smart enable
diagnostic bootup level minimal
spanning-tree extend system-id
redundancy
mode none
interface GigabitEthernet0/0/0
ip address 192.168.1.2 255.255.255.0
ip ospf 10 area 1
negotiation auto
ipv6 address 2001:DB8:ACAD:B::2/64
ipv6 enable
ipv6 ospf 10 area 1
interface GigabitEthernet0/0/1
ip address 192.168.2.1 255.255.255.0
ip ospf 10 area 1
negotiation auto
ipv6 address 2001:DB8:ACAD:C::1/64
ipv6 enable
ipv6 ospf 10 area 1
interface GigabitEthernet0/2/0
```

```
no ip address
negotiation auto
interface GigabitEthernet0/2/1
no ip address
negotiation auto
interface GigabitEthernet0
vrf forwarding Mgmt-intf
no ip address
negotiation auto
router ospf 10
router-id 6.6.6.6
ip forward-protocol nd
no ip http server
ip http secure-server
ipv6 router ospf 10
router-id 6.6.6.6
control-plane
line con 0
transport input none
stopbits 1
line aux 0
stopbits 1
line vty 0 4
login
end
```

Router2:

```
Current configuration : 1569 bytes
Last configuration change at 22:00:46 UTC Mon Sep 30 2024
version 15.5
service timestamps debug datetime msec
service timestamps log datetime msec
no platform punt-keepalive disable-kernel-core
hostname Router2
boot-start-marker
boot-end-marker
vrf definition Mgmt-intf
address-family ipv4
exit-address-family
address-family ipv6
exit-address-family
no aaa new-model
ipv6 unicast-routing
subscriber templating
multilink bundle-name authenticated
license udi pid ISR4321/K9 sn FDO214420G7
spanning-tree extend system-id
redundancy
mode none
vlan internal allocation policy ascending
interface GigabitEthernet0/0/0
```

```
ip address 192.168.2.2 255.255.255.0
ip ospf 10 area 1
negotiation auto
ipv6 address 2001:DB8:ACAD:C::2/64
ipv6 enable
ipv6 ospf 10 area 1
interface GigabitEthernet0/0/1
ip address 192.168.3.1 255.255.255.0
ip ospf 10 area 1
negotiation auto
ipv6 address 2001:DB8:ACAD:D::1/64
ipv6 enable
ipv6 ospf 10 area 1
interface Serial0/1/0
no ip address
shutdown
interface Serial0/1/1
no ip address
shutdown
interface GigabitEthernet0
vrf forwarding Mgmt-intf
no ip address
shutdown
negotiation auto
interface Vlan1
no ip address
shutdown
router ospf 10
router-id 3.3.3.3
ip forward-protocol nd
no ip http server
no ip http secure-server
ip tftp source-interface GigabitEthernet0
ipv6 router ospf 10
router-id 6.6.6.6
control-plane
line con 0
stopbits 1
line aux 0
stopbits 1
line vty 0 4
login
end
```

Multilayer Switch:

```
Current configuration : 3723 bytes
Last configuration change at 00:04:31 UTC Mon Mar 1 1993
version 12.2
no service pad
service timestamps debug datetime msec
service timestamps log datetime msec
```

```
no service password-encryption
hostname S1
boot-start-marker
boot-end-marker
no aaa new-model
system mtu routing 1500
ip routing
ipv6 unicast-routing
spanning-tree mode pvst
spanning-tree extend system-id
vlan internal allocation policy ascending
interface FastEthernet1/0/1
no switchport
ip address 192.168.6.1 255.255.255.0
ipv6 address FE80::1 link-local
ipv6 address 2001:DB8:ACAD:A1::1/64
ipv6 ospf 10 area 0
interface FastEthernet1/0/2
interface FastEthernet1/0/3
interface FastEthernet1/0/4
interface FastEthernet1/0/5
interface FastEthernet1/0/6
interface FastEthernet1/0/7
interface FastEthernet1/0/8
interface FastEthernet1/0/9
interface FastEthernet1/0/10
interface FastEthernet1/0/11
interface FastEthernet1/0/12
interface FastEthernet1/0/13
interface FastEthernet1/0/14
interface FastEthernet1/0/15
interface FastEthernet1/0/16
interface FastEthernet1/0/17
interface FastEthernet1/0/18
interface FastEthernet1/0/19
interface FastEthernet1/0/20
interface FastEthernet1/0/21
interface FastEthernet1/0/22
interface FastEthernet1/0/23
interface FastEthernet1/0/24
interface GigabitEthernet1/0/1
no switchport
ip address 192.168.0.2 255.255.255.0
ipv6 address FE80::2 link-local
ipv6 address 2001:DB8:ACAD:A::2/64
ipv6 ospf 10 area 0
interface GigabitEthernet1/0/2
no switchport
ip address 192.168.4.1 255.255.255.0
ipv6 address FE80::1 link-local
ipv6 address 2001:DB8:ACAD:E::1/64
ipv6 ospf 10 area 2
```

```
interface GigabitEthernet1/1/1
interface GigabitEthernet1/1/2
interface Vlan1
no ip address
shutdown
router ospf 10
router-id 5.5.5.5
network 192.168.0.0 0.0.0.255 area 0
network 192.168.4.0 0.0.0.255 area 2
network 192.168.6.0 0.0.0.255 area 0
ip http server
ip http secure-server
logging esm config
ipv6 router ospf 10
line con 0
line vty 0 4
login
line vty 5 15
login
end
```

Router3:

```
Current configuration : 1605 bytes
version 15.5
service timestamps debug datetime msec
service timestamps log datetime msec
no platform punt-keepalive disable-kernel-core
hostname Router3
boot-start-marker
boot-end-marker
vrf definition Mgmt-intf
address-family ipv4
exit-address-family
address-family ipv6
exit-address-family
no aaa new-model
ipv6 unicast-routing
subscriber templating
multilink bundle-name authenticated
license udi pid ISR4321/K9 sn FDO211216BL
spanning-tree extend system-id
redundancy
mode none
vlan internal allocation policy ascending
interface GigabitEthernet0/0/0
ip address 192.168.4.2 255.255.255.0
ip ospf 10 area 2
```

```
negotiation auto
ipv6 address 2001:DB8:ACAD:E::2/64
ipv6 enable
ipv6 ospf 10 area 2
interface GigabitEthernet0/0/1
ip address 192.168.5.1 255.255.255.0
ip ospf 10 area 2
negotiation auto
ipv6 address 2001:DB8:ACAD:F::1/64
ipv6 enable
ipv6 ospf 10 area 2
interface Serial0/1/0
no ip address
interface Serial0/1/1
no ip address
interface GigabitEthernet0
vrf forwarding Mgmt-intf
no ip address
negotiation auto
interface Vlan1
no ip address
router ospf 10
router-id 2.2.2.2
network 192.168.4.0 0.0.0.255 area 2
network 192.168.5.0 0.0.0.255 area 2
ip forward-protocol nd
no ip http server
no ip http secure-server
ip tftp source-interface GigabitEthernet0
ipv6 router ospf 10
router-id 2.2.2.2
control-plane
line con 0
stopbits 1
line aux 0
stopbits 1
line vty 0 4
login
end
```

Router4:

```
Current configuration : 1526 bytes
version 15.5
service timestamps debug datetime msec
service timestamps log datetime msec
```

```
no platform punt-keepalive disable-kernel-core
hostname Router4
boot-start-marker
boot-end-marker
vrf definition Mgmt-intf
address-family ipv4
exit-address-family
address-family ipv6
exit-address-family
no aaa new-model
ipv6 unicast-routing
subscriber templating
multilink bundle-name authenticated
license udi pid ISR4321/K9 sn FDO214421CF
spanning-tree extend system-id
redundancy
mode none
vlan internal allocation policy ascending
interface Loopback0
no ip address
interface GigabitEthernet0/0/0
ip address 192.168.5.2 255.255.255.0
ip ospf 10 area 2
negotiation auto
ipv6 address 2001:DB8:ACAD:F::2/64
ipv6 enable
ipv6 ospf 10 area 2
interface GigabitEthernet0/0/1
no ip address
negotiation auto
ipv6 enable
ipv6 ospf 10 area 2
interface Serial0/1/0
no ip address
interface Serial0/1/1
no ip address
interface GigabitEthernet0
vrf forwarding Mgmt-intf
no ip address
negotiation auto
interface Vlan1
no ip address
router ospf 10
router-id 1.1.1.1
network 192.168.5.0 0.0.0.255 area 2
ip forward-protocol nd
no ip http server
no ip http secure-server
ip tftp source-interface GigabitEthernet0
ipv6 router ospf 10
router-id 1.1.1.1
control-plane
```

```
line con 0
stopbits 1
line aux 0
stopbits 1
line vty 0 4
login
end
```

```
Gateway of last resort is not set

      192.168.0.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.0.0/24 is directly connected, GigabitEthernet1/0/1
L        192.168.0.2/32 is directly connected, GigabitEthernet1/0/1
O IA  192.168.1.0/24 [110/2] via 192.168.0.1, 00:13:14, GigabitEthernet1/0/1
O IA  192.168.2.0/24 [110/3] via 192.168.0.1, 00:05:46, GigabitEthernet1/0/1
O IA  192.168.3.0/24 [110/4] via 192.168.0.1, 00:00:52, GigabitEthernet1/0/1
      192.168.4.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.4.0/24 is directly connected, GigabitEthernet1/0/2
L        192.168.4.1/32 is directly connected, GigabitEthernet1/0/2
O     192.168.5.0/24 [110/2] via 192.168.4.2, 00:15:51, GigabitEthernet1/0/2
S1#
```

```
S1#show ipv6 route
IPv6 Routing Table - default - 9 entries
Codes: C - Connected, L - Local, S - Static, U - Per-user Static route
       B - BGP, R - RIP, D - EIGRP, EX - EIGRP external
       ND - Neighbor Discovery
       O - OSPF Intra, OI - OSPF Inter, OE1 - OSPF ext 1, OE2 - OSPF ext 2
       ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2
C   2001:DB8:ACAD:A::/64 [0/0]
     via GigabitEthernet1/0/1, directly connected
L   2001:DB8:ACAD:A::2/128 [0/0]
     via GigabitEthernet1/0/1, receive
OI  2001:DB8:ACAD:B::/64 [110/2]
     via FE80::1, GigabitEthernet1/0/1
OI  2001:DB8:ACAD:C::/64 [110/3]
     via FE80::1, GigabitEthernet1/0/1
OI  2001:DB8:ACAD:D::/64 [110/4]
     via FE80::1, GigabitEthernet1/0/1
C   2001:DB8:ACAD:E::/64 [0/0]
     via GigabitEthernet1/0/2, directly connected
L   2001:DB8:ACAD:E::1/128 [0/0]
     via GigabitEthernet1/0/2, receive
O   2001:DB8:ACAD:F::/64 [110/2]
     via FE80::2F8:2CFF:FE7F:7190, GigabitEthernet1/0/2
L   FF00::/8 [0/0]
     via Null0, receive
```

```
Router4#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

O IA  192.168.0.0/24 [110/3] via 192.168.5.1, 00:17:28, GigabitEthernet0/0/0
O IA  192.168.1.0/24 [110/4] via 192.168.5.1, 00:15:02, GigabitEthernet0/0/0
O IA  192.168.2.0/24 [110/5] via 192.168.5.1, 00:07:33, GigabitEthernet0/0/0
O IA  192.168.3.0/24 [110/6] via 192.168.5.1, 00:02:39, GigabitEthernet0/0/0
O     192.168.4.0/24 [110/2] via 192.168.5.1, 00:51:26, GigabitEthernet0/0/0
      192.168.5.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.5.0/24 is directly connected, GigabitEthernet0/0/0
L        192.168.5.2/32 is directly connected, GigabitEthernet0/0/0
Router4#show ipv6 route
IPv6 Routing Table - default - 8 entries
Codes: C - Connected, L - Local, S - Static, U - Per-user Static route
       B - BGP, R - RIP, I1 - ISIS L1, I2 - ISIS L2
       IA - ISIS interarea, IS - ISIS summary, D - EIGRP, EX - EIGRP external
       ND - ND Default, NDp - ND Prefix, DCE - Destination, NDr - Redirect
       O - OSPF Intra, OI - OSPF Inter, OE1 - OSPF ext 1, OE2 - OSPF ext 2
       ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2, a - Application
OI  2001:DB8:ACAD:A::/64 [110/3]
     via FE80::2F8:2CFF:FE7F:7191, GigabitEthernet0/0/0
OI  2001:DB8:ACAD:B::/64 [110/4]
     via FE80::2F8:2CFF:FE7F:7191, GigabitEthernet0/0/0
OI  2001:DB8:ACAD:C::/64 [110/5]
     via FE80::2F8:2CFF:FE7F:7191, GigabitEthernet0/0/0
OI  2001:DB8:ACAD:D::/64 [110/6]
     via FE80::2F8:2CFF:FE7F:7191, GigabitEthernet0/0/0
O   2001:DB8:ACAD:E::/64 [110/2]
     via FE80::2F8:2CFF:FE7F:7191, GigabitEthernet0/0/0
C   2001:DB8:ACAD:F::/64 [0/0]
     via GigabitEthernet0/0/0, directly connected
L   2001:DB8:ACAD:F::2/128 [0/0]
     via GigabitEthernet0/0/0, receive
L   FF00::/8 [0/0]
     via Null0, receive
Router4#
```

```
Gateway of last resort is not set

O IA  192.168.0.0/24 [110/2] via 192.168.4.1, 00:16:33, GigabitEthernet0/0/0
O IA  192.168.1.0/24 [110/3] via 192.168.4.1, 00:14:07, GigabitEthernet0/0/0
O IA  192.168.2.0/24 [110/4] via 192.168.4.1, 00:06:38, GigabitEthernet0/0/0
O IA  192.168.3.0/24 [110/5] via 192.168.4.1, 00:01:45, GigabitEthernet0/0/0
      192.168.4.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.4.0/24 is directly connected, GigabitEthernet0/0/0
L        192.168.4.2/32 is directly connected, GigabitEthernet0/0/0
      192.168.5.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.5.0/24 is directly connected, GigabitEthernet0/0/1
L        192.168.5.1/32 is directly connected, GigabitEthernet0/0/1
Router3#
```

```
Router3#show ipv6 route
IPv6 Routing Table - default - 9 entries
Codes: C - Connected, L - Local, S - Static, U - Per-user Static route
       B - BGP, R - RIP, I1 - ISIS L1, I2 - ISIS L2
       IA - ISIS interarea, IS - ISIS summary, D - EIGRP, EX - EIGRP external
       ND - ND Default, NDp - ND Prefix, DCE - Destination, NDr - Redirect
       O - OSPF Intra, OI - OSPF Inter, OE1 - OSPF ext 1, OE2 - OSPF ext 2
       ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2, a - Application
OI  2001:DB8:ACAD:A::/64 [110/2]
     via FE80::1, GigabitEthernet0/0/0
OI  2001:DB8:ACAD:B::/64 [110/3]
     via FE80::1, GigabitEthernet0/0/0
OI  2001:DB8:ACAD:C::/64 [110/4]
     via FE80::1, GigabitEthernet0/0/0
OI  2001:DB8:ACAD:D::/64 [110/5]
     via FE80::1, GigabitEthernet0/0/0
C   2001:DB8:ACAD:E::/64 [0/0]
     via GigabitEthernet0/0/0, directly connected
L   2001:DB8:ACAD:E::2/128 [0/0]
     via GigabitEthernet0/0/0, receive
C   2001:DB8:ACAD:F::/64 [0/0]
     via GigabitEthernet0/0/1, directly connected
L   2001:DB8:ACAD:F::1/128 [0/0]
     via GigabitEthernet0/0/1, receive
L   FF00::/8 [0/0]
     via Null0, receive
Router3#
```

```
Router0#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

      192.168.0.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.0.0/24 is directly connected, GigabitEthernet0/0/0
L        192.168.0.1/32 is directly connected, GigabitEthernet0/0/0
      192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.1.0/24 is directly connected, GigabitEthernet0/0/1
L        192.168.1.1/32 is directly connected, GigabitEthernet0/0/1
O     192.168.2.0/24 [110/2] via 192.168.1.2, 00:17:52, GigabitEthernet0/0/1
O     192.168.3.0/24 [110/3] via 192.168.1.2, 00:12:58, GigabitEthernet0/0/1
O IA  192.168.4.0/24 [110/2] via 192.168.0.2, 00:25:25, GigabitEthernet0/0/0
O IA  192.168.5.0/24 [110/3] via 192.168.0.2, 00:25:25, GigabitEthernet0/0/0
Router0#
```

```
Router1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

O IA  192.168.0.0/24 [110/2] via 192.168.1.1, 00:22:37, GigabitEthernet0/0/0
      192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.1.0/24 is directly connected, GigabitEthernet0/0/0
L        192.168.1.2/32 is directly connected, GigabitEthernet0/0/0
      192.168.2.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.2.0/24 is directly connected, GigabitEthernet0/0/1
L        192.168.2.1/32 is directly connected, GigabitEthernet0/0/1
O     192.168.3.0/24 [110/2] via 192.168.2.2, 00:14:01, GigabitEthernet0/0/1
O IA  192.168.4.0/24 [110/3] via 192.168.1.1, 00:22:37, GigabitEthernet0/0/0
O IA  192.168.5.0/24 [110/4] via 192.168.1.1, 00:22:37, GigabitEthernet0/0/0
Router1#
```

```
Router2#sho
*Sep 30 22:15:09.196: %SYS-5-CONFIG_I: Configured from console by consolew ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

O IA  192.168.0.0/24 [110/3] via 192.168.2.1, 00:19:17, GigabitEthernet0/0/0
O     192.168.1.0/24 [110/2] via 192.168.2.1, 00:19:17, GigabitEthernet0/0/0
      192.168.2.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.2.0/24 is directly connected, GigabitEthernet0/0/0
L        192.168.2.2/32 is directly connected, GigabitEthernet0/0/0
      192.168.3.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.3.0/24 is directly connected, GigabitEthernet0/0/1
L        192.168.3.1/32 is directly connected, GigabitEthernet0/0/1
O IA  192.168.4.0/24 [110/4] via 192.168.2.1, 00:19:17, GigabitEthernet0/0/0
O IA  192.168.5.0/24 [110/5] via 192.168.2.1, 00:19:17, GigabitEthernet0/0/0
Router2#
```

Problems:

My first problem was struggling to figure out how to configure the multi-layer switch with IP addresses for OSPF and routing. This was found to be the missing `sdm prefer dual-ipv4-and-ipv6 default` command which enables layer 3 capabilities on the switch.

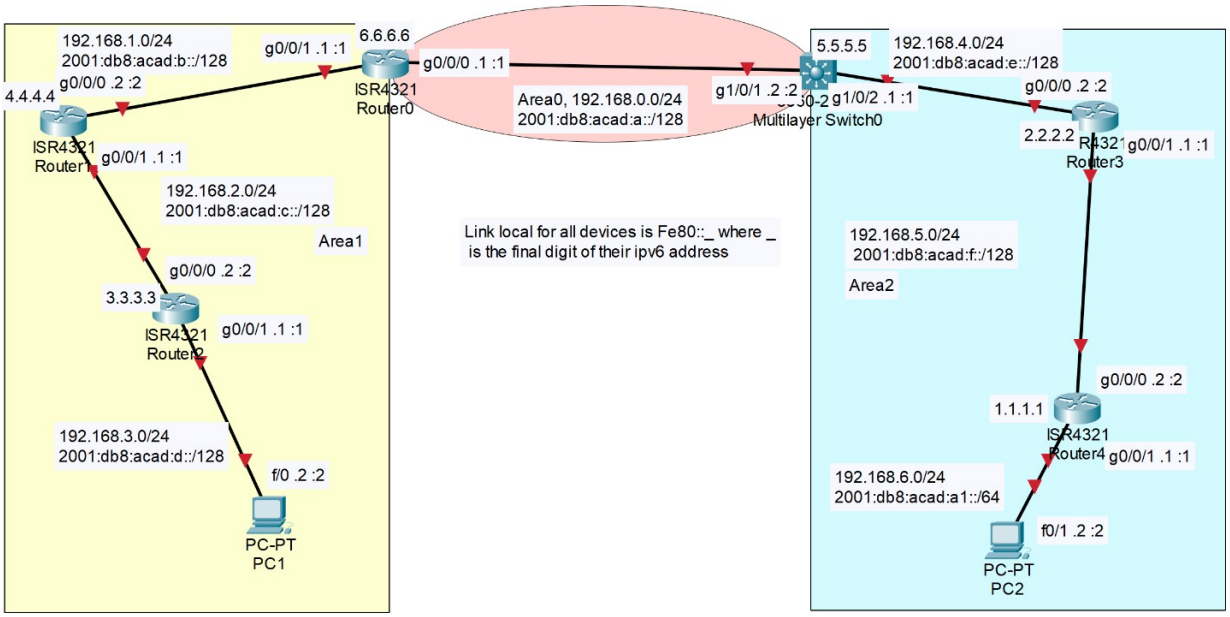
My second problem happened when I had router to router connectivity across the network but could not ping end-to-end, the problem was found to be the computers being connected to a Wi-Fi network, and packets being sent through the Wi-Fi network and not through the wired ethernet connections, I disconnected from the Wi-Fi network and end-to-end connectivity was achieved.

Conclusion:

The lab was overall a success in refreshing my understanding of OSPF networks, and more crucially of multi-area OSPF networks which have become a cornerstone of the industry, for their efficiency, simplicity, and high performance. With small hiccups in network connectivity and multi-layer switch connectivity

Diagram and Addressing:

Device name	ospf area	Interface	IPv6 add	IPv4 add	Subnet Mask	link local
R0	Area 0	G0/0/0	2001:db8:acad:a::1/64	192.168.0.1	255.255.255.0	Fe80::1
S1	Area 0	G1/0/1	2001:db8:acad:a::2/64	192.168.0.2	255.255.255.0	Fe80::2
R0	Area 1	G0/0/1	2001:db8:acad:b::1/64	192.168.1.1	255.255.255.0	Fe80::1
R1	Area 1	G0/0/0	2001:db8:acad:b::2/64	192.168.1.2	255.255.255.0	Fe80::2
R1	Area 1	G0/0/1	2001:db8:acad:c::1/64	192.168.2.1	255.255.255.0	Fe80::1
R2	Area 1	G0/0/0	2001:db8:acad:c::2/64	192.168.2.2	255.255.255.0	Fe80::2
R2	Area 1	G0/0/1	2001:db8:acad:d::1/64	192.168.3.1	255.255.255.0	Fe80::1
PC1	Area 1	F/0	2001:db8:acad:d::2/64	192.168.3.2	255.255.255.0	Fe80::2
S1	Area 2	G1/0/2	2001:db8:acad:e::1/64	192.168.4.1	255.255.255.0	Fe80::1
R3	Area 2	G0/0/0	2001:db8:acad:e::2/64	192.168.4.2	255.255.255.0	Fe80::2
R3	Area 2	G0/0/1	2001:db8:acad:f::1/64	192.168.5.1	255.255.255.0	Fe80::1
R4	Area 2	G0/0/0	2001:db8:acad:f::2/64	192.168.5.2	255.255.255.0	Fe80::2
R4	Area 2	G0/0/1	2001:db8:acad:a1::1/64	192.168.6.1	255.255.255.0	Fe80::1
PC2	Area 2	f/0	2001:db8:acad:a1::2/64	192.168.6.2	255.255.255.0	Fe80::2



Signoff

This confirms that I, Alex Huhdanpaa completed the

Multi-Area OSPF configuration lab to a satisfactory standard

_____X